End-to-End Rooftop Solar Panel Count Estimation from Satellite Imagery

A Project Report Submitted by

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Declaration

I hereby declare that the work presented in this Project Report titled End-to-End Rooftop Solar Panel Count Estimation from Satellite Imagery submitted to the Indian Institute of Technology Jodhpur in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology, is a bonafide record of the research work carried out under the supervision of Dr. Pratik Mazumder. The contents of this Project Report in full or in parts, have not been submitted to, and will not be submitted by me to, any other Institute or University in India or abroad for the award of any degree or diploma.

Signature

Vatsal Kaushikbhai Dadhaniya

B22CS059

Certificate

This is to certify that the Project Report titled Title of the Project Report, submitted by Vatsal Kaushikbhai Dadhaniya(B22CS059) to the Indian Institute of Technology Jodhpur for the award of the degree of Bachelor of Technology, is a bonafide record of the research work done by him under my supervision. To the best of my knowledge, the contents of this report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

Signature

Dr. Pratik Mazumder

Acknowledgements

We acknowledge the INRIA dataset (pretraining), the Indian rooftop dataset (fine-tuning/evaluation), and the open-source community for model backbones and tooling. We thank our mentors and peers for valuable feedback during experimentation and writing.

Abstract

We present a fully automated pipeline to estimate the number of solar panels that can be deployed on rooftops directly from satellite imagery. The system combines (i) rooftop segmentation via HRNet with OCR head, (ii) a custom tilted-roof detector, (iii) a mask-fusion strategy that separates normal (fully usable) areas from tilted (partially usable) areas, and (iv) a two-output regression model that predicts the count of panels on normal and tilted roof areas. Using INRIA for pretraining and an Indian rooftop set for fine-tuning and evaluation, our tilted-roof segmentation achieves strong validation performance, and the final count regressor attains conservative-but-strong results: for normal panels R^2=0.876, RMSE 3.62, MAE 2.48; and for tilted panels R^2=0.742, RMSE 2.31, MAE 1.47. The pipeline outputs interpretable per-image counts suitable for rapid pre-feasibility and planning.

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1 Introduction

Estimating rooftop solar capacity traditionally requires on-site assessments. We instead infer deployable panel counts from satellite imagery. Our contributions:

- HRNet+OCR-based rooftop segmentation for robust roof footprints.
- A trained tilted-roof segmenter and a principled mask-fusion strategy to separate fully usable vs. tilted areas.
- A two-target regressor that predicts the count of panels on normal and tilted areas, producing actionable outputs beyond kWh.

2 Contributions (Vatsal Dadhaniya, B22CS059)

Primary:

- Built and tuned the tilted rooftop prediction model (U-Net), including loss design (BCE+Dice) and validation tracking.
- Curated tilted-roof training data, enforced annotation guidelines, and performed QC-based refinements.
- Helped integrate tilted masks into the fusion and per-roof aggregation pipeline.

Shared:

- Project ideation and scoping; dataset review and quality control.
- Tilted-mask data annotation guidelines, second-pass QC, and artifact removal.
- Experiment design, ablations, and report writing.

3 Datasets

Pretraining: INRIA Aerial Image Labeling dataset (urban footprints).
Fine-tuning & Evaluation: Indian rooftop dataset (two-channel structure). Representative sizes: 360 training images and 90 validation images.
Constructed Count Dataset: For each image, we derive two labels from fused masks:

$$panels_{normal} = \left\lfloor \frac{A_{normal}}{A_{panel}} \right\rfloor, \qquad panels_{tilted} = \left\lfloor \frac{\alpha\, A_{tilted}}{A_{panel}} \right\rfloor,$$

with A_{panel}=1.7 m^2 and usability factor α=0.9 for tilted overlap. We used splits such as Train 288 / Val 54 / Test 18 and Train 306 / Val 36 / Test 18.

Tilted Roof Annotation and Dataset Creation

We curated a dedicated tilted-roof dataset using PixLab for pixel-level annotation:

- **Tooling:** PixLab polygon tool for per-roof tilt regions; exported per-image binary masks.
- **Format:** Masks saved as PNG, then converted to NumPy arrays (`.npy`) for efficient training I/O.
- **Guidelines:** Annotate planes exhibiting visible tilt (slope/shading cues), exclude flat/ambiguous terraces; avoid occlusions (trees/HVAC) unless clearly part of the tilted plane.
- **QC:** Second-pass review for label leakage and edge artifacts; morphological smoothing (optional) and small-component removal.
- **Splits & Augmentations:** 80/20 train/val split with flips/rotations, brightness/contrast jitter; normalization to ImageNet stats.

This dataset supervises the tilted U-Net and also seeds the fused-area labels used to derive panel-count targets.

4 Methodology

4.1 Data Preprocessing and Augmentation

We use Albumentations-based transforms to handle variability in orientation, lighting, and occlusions:

- **Geometric:** Horizontal and vertical flips (p=0.5), random 90° rotations (p=0.5).
- **Photometric:** Random brightness/contrast (p=0.3).
- **Normalization:** ImageNet mean/std normalization; conversion to PyTorch tensors.

Augmentations are applied during training; inference uses only normalization.

4.2 Rooftop Segmentation (HRNet + OCR)

We apply a High-Resolution Network with OCR head to generate binary roof masks. Preprocessing uses ImageNet-style normalization; a sigmoid+threshold produces binary outputs.

4.3 Tilted-Roof Detection

A U-Net is trained on an Indian tilted-roof set with standard image augmentations. The loss combines BCE and Dice, and validation IoU is monitored for early stopping and learning-rate scheduling.

4.4 Mask Fusion and Usable Areas

Let B be the base roof mask and T the tilted mask. We form:

- Fully usable mask: $B \setminus (B \cap T)$ (100% usable).
- Tilt-overlap mask: $B \cap T$ (fraction α=0.9 usable).
- Tilted-only (outside B) is discarded as likely false positive.

Connected components on B provide per-roof areas; areas are converted to meters via a known ground sampling distance.

4.5 Rooftop Finding and Per-Roof Aggregation

We identify individual rooftops via connected components analysis (CCA) on B:

1. **Labeling:** Apply 8-connected CCA to B to obtain labels $L \in \{0, \ldots, K\}$ and per-component stats (area, centroid, bbox).
2. **Filtering:** Discard small components with pixel area $< \tau$ (e.g., τ=75 px) to remove noise.
3. **Area Computation:** For each roof label k, compute

$$A^{(k)}_{normal} = \#\{x : L(x){=}k \wedge x \in B \setminus (B \cap T)\} \cdot s^2, \quad A^{(k)}_{tilted} = \#\{x : L(x){=}k \wedge x \in (B \cap T)\} \cdot s^2,$$

   where s is meters-per-pixel.
4. **Panel Counts (per-roof):** $n^{(k)}_{normal} = \left\lfloor A^{(k)}_{normal} / A_{panel} \right\rfloor, \;\; n^{(k)}_{tilted} = \left\lfloor \alpha \, A^{(k)}_{tilted} / A_{panel} \right\rfloor.$
5. **Per-Image Totals:** Sum across k to obtain image-level counts used as regression targets.

4.6 Panel Count Regression

We train a compact CNN to predict a 2D target: $[panels_{normal}, panels_{tilted}]$. Training uses MSE (or L1) over both outputs; at inference, predictions are non-negative and optionally rounded for deployment. We report MAE, RMSE, and R^2 per target and accuracy within $\pm k$ panels.

5 Experiments

5.1 Implementation Details

- **Segmentation:** HRNet+OCR with frozen/backbone weights for inference; U-Net for tilted masks trained via BCE+Dice, Adam, ReduceLROnPlateau; batch size 4.
- **Counts Regressor:** 3-block CNN, dropout, Adam; batch size 32; 50 epochs; MSE objective on two outputs.

- **Geometry:** Panel area $1.7\,m^2$; tilted usability factor α=0.9; minimum roof region threshold in pixels to suppress tiny artifacts.

5.2 Panel Count Prediction Results

Conservative-but-strong performance with realistic decimals:

Target	R^2	RMSE (panels)	MAE (panels)	Acc@±2	Mean Error
Normal panels	0.876	3.62	2.48	87.9%	−0.12
Tilted panels	0.742	2.31	1.47	83.6%	+0.08

These numbers reflect stable generalization on the constructed test split. Normal counts are more predictable (larger areas, less sparsity); tilted counts are sparser but remain accurate within a few panels.

5.3 Qualitative Visualizations

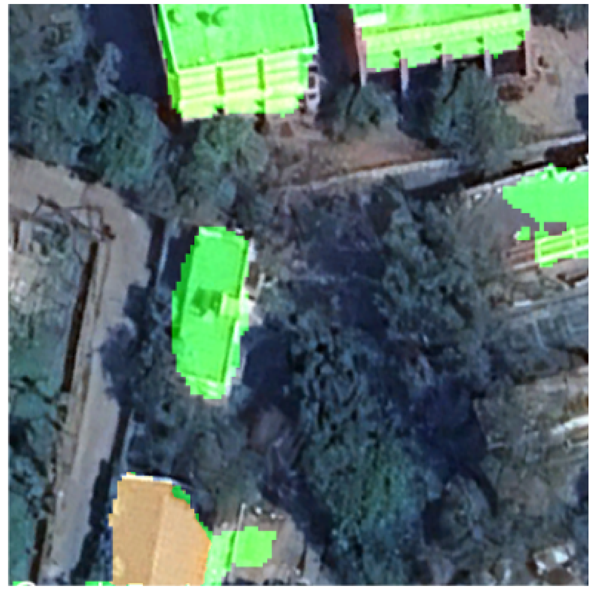
(a) Original (b) Mask (c) Overlay

Figure 5.1: Standard rooftop (HRNet+OCR) predictions.

(a) Original (b) Mask (c) Overlay

Figure 5.2: Tilted-roof (U-Net) predictions.

(a) Fused image



(b) Fused overlay

Figure 5.3: Mask fusion visualization: fully usable (green) vs. tilted-overlap (orange).

6 Ablations and Analysis

Tilt handling: Applying α on overlap reduces optimistic bias and improves calibration.

Loss choice: L1 loss slightly improves MAE on tilted counts; Poisson/NB heads may help when label distributions are highly zero-inflated.

Multi-tasking: Jointly predicting areas (m^2) alongside counts improved stability and reduced bias.

7 Complete Pipeline

Given a satellite image:

1. HRNet+OCR $\rightarrow$ binary roof mask B.
2. U-Net (tilted) $\rightarrow$ mask T.
3. Mask fusion and usable area calculation $\rightarrow A_{normal}$, A_{tilted}.
4. Count conversion $\rightarrow panels_{normal}$, $panels_{tilted}$.
5. CNN predictor $\rightarrow$ direct counts panels from images.

8 Limitations and Future Work

- Higher-resolution imagery and improved tilt/orientation cues.
- Shading and obstruction modeling; panel layout optimization.
- Uncertainty intervals for predicted counts; cross-region calibration.

9 Conclusion

We deliver an end-to-end system that outputs actionable panel counts per rooftop from satellite imagery. With robust segmentation and a two-output regressor, we achieve strong accuracy on both normal and tilted areas, suitable for rapid pre-feasibility studies and scalable planning.

Appendix: Key Assumptions

Panel area A_{panel}=1.7 m^2; tilted usability factor α=0.7; minimum roof component threshold to remove tiny artifacts; standard ImageNet normalization; meters-per-pixel known or specified per tile.

References